

Earbud-Size FM RADIO

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In the June issue, the construction of a 'Miniature Watch-Style FM Radio' delivering stereo FM output was described. Building on that concept, the next objective was to create an even smaller FM radio compact enough to fit inside an earbud. However, the complete design proved too complex for a single article and is therefore being divided into parts.

This part, covering 'Earbud-size FM radio,' presents a simplified version—a basic FM radio embedded in an earbud with limited functionality. It allows listening to a single, preset FM channel configured on the main FM integrated circuit (IC). Alternatively, by presetting the IC to a specific frequency, the earbud can serve as a long-range FM spy device, receiving audio transmitted on that frequency.

The next part in this series of FM radio articles will introduce a tiny SMD-based microcontroller, such as the ATtiny85 or one of the world's smallest MCUs from Texas Instruments (like the MSPM0C1104: https://www.ti.com/product/MSPM0C1104?hqs=epd-msp-procbr-mspm0_small-pr-pf-ew25-wwe), enabling complete programmatic control of the FM radio.

The current version is a fixed-frequency FM radio earbud. The FM chip must be programmed with the desired frequency before it can be used. To allow for reconfiguration, two conductive contact points have been added to the earbud, enabling frequency adjustments when needed. For spy applications, the frequency can be matched to that of a compatible FM transmitter.

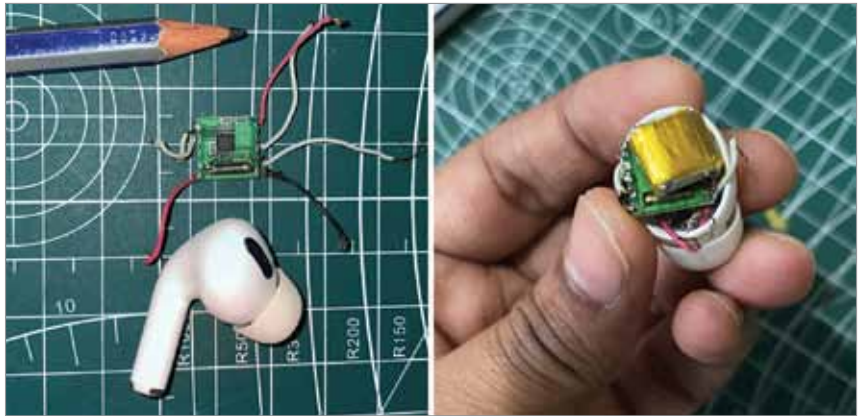


Fig. 1: Author's prototype with RDA580 module

BILL OF MATERIALS

Component	Part Number	Quantity
A 30 to 50mAh battery used in TWS (true wireless stereo) earbuds	A compact Li-Po battery such as the EEMB LP401012, D10 KP-581013X, or a similar model	1
A 16-ohm, 7mm diameter speaker is typically used in earbuds	MACGARSEN (component manufacturer)	1
IndusBoard Coin/Arduino	Coin V2 (for programming FM CHIP)	1
RDA5807M FM Stereo Radio Module	RDA5807M	1
FM radio audio transmitter	FM wireless microphone transmitter module	1

Fig. 1 shows the prototype featuring the RDA580 module. The components required to build the device are listed in the Bill of Materials table.

Circuit diagram and programming

Three different circuits are involved in this device. The first is the circuit for the RDA5807M radio IC, which must be programmed to tune to a specific FM frequency. Fig. 2 shows the circuit for programming the board to configure the FM radio frequency and volume.

This initial programming circuit is straightforward: the IndusBoard Coin is connected to the FM chip via the I²C pins (SCL and SDA), along

with power connections to GND and 3.3V. Once programmed and configured to the desired volume and frequency, the FM radio functions independently.

The next circuit is the earbud circuit, where a TWS earbud battery powers the FM chip and a 16-ohm, 7mm earbud speaker is connected to the audio output of the RDA5807M chip. All components are then installed into a standard AirPods-style case. Fig. 3 shows the earbud circuit.

For applications requiring an FM spy device, the circuit for the FM audio transmitter is also provided. Fig. 4 shows the FM radio transmitter circuit.